ring is formed. In addition, vascular bundles are scattered or found in a ring in the pith or medullary region. These are called **medullary vascular bundles**. Example: *Boerhaavia.*

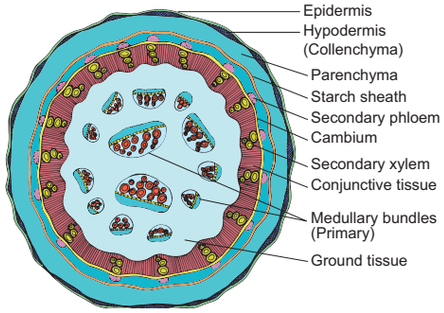


Figure 10.20: Transverse section (T.S) of *Boerhaavia* stem (diagrammatic) showing anomalous secondary growth

10.3.6 Presence of cortical bundles along with normal cambial activity

In this type, a ring of vascular bundles are found, from which a normal cambial ring is formed. In addition, in the cortical region additional vascular bundles are found. Example: *Nyctanthes.*

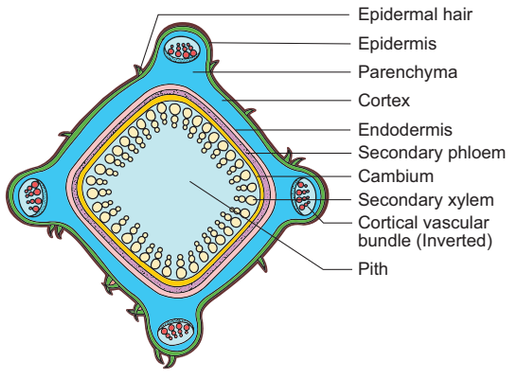


Figure 10.21: Transverse section (T.S) of *Nyctanthes* stem (diagrammatic) showing anomalous secondary growth

10.3.7 Intraxylary or Internal Phloem

The cambium usually forms primary phloem on the outer side. In some plants,it forms the primary phloem on the inner side along with xylem towards pith. This is called **intraxylary** or **internal Phloem**. Example: *Solanum tuberosum*



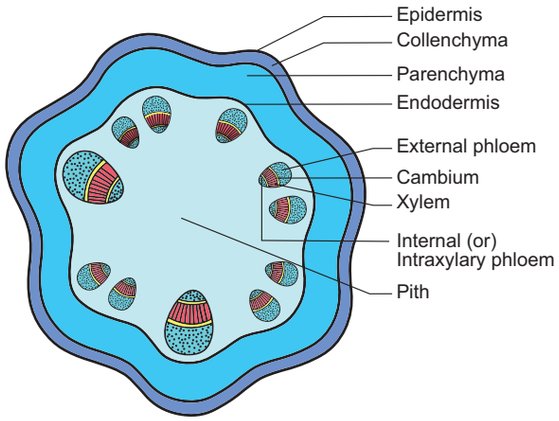


Figure 10.22: Transverse section (T.S) of *Solanum tuberosum* stem (diagrammatic) showing anomalous secondary growth

10.3.8 Secondary Growth in Monocot

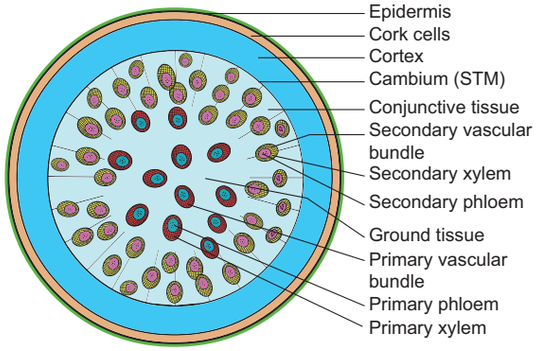


Figure 10.23: Transverse section of *Dracaena* stem (diagrammatic) showing anomalous secondary growth

In some monocotyledons, the stems exhibit secondary growth. Example: *Dracaena*. Here, a secondary thickening meristems (STM) originate from the ground tissue outside of the vascular bundles. STM cuts off cells on the inside from which additional vascular bundles are formed along with parenchyma cells in between the vascular bundles. The STM consists of only one type of cells unlike normal vascular cambium. STM does not produce continuous xylem on the inside but only vascular bundle with xylem and phloem. Also no phloem is formed on the outer side.

10.3.9 Anomalous primary growth

I) Absence of vessels in the xylem.

Usually, vessels are found in the xylem of angiosperms, whereas some plants are

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